Lab 5 Part 2 – Mitigating Against STP Manipulation & CAM Table Attacks

Objectives

- Assign the Core switch as the root bridge.
- Secure spanning-tree parameters to prevent STP manipulation attacks.
- Enable port security to prevent CAM table overflow attacks.

Background / Scenario

There have been a number of attacks on the network recently. For this reason, the network administrator has assigned you the task of configuring Layer 2 security.

For optimum performance and security, the administrator would like to ensure that the root bridge is the 3560 Core switch. To prevent spanning-tree manipulation attacks, the administrator wants to ensure that the STP parameters are secure. To prevent against CAM table overflow attacks, the network administrator has decided to configure port security to limit the number of MAC addresses each switch port can learn. If the number of MAC addresses exceeds the set limit, the administrator would like the port to be shutdown.

All switch devices have been preconfigured with the following:

- Enable password: **ciscoenpa55**
- Console password: **ciscoconpa55**
- SSH username and password: **SSHadmin** / **ciscosshpa55**

Part 1: Configure Root Bridge

Step 1: Determine the current root bridge.

On **Core**, **SW-1** and **SW-2** issue the **show spanning-tree** command to determine the current root bridge, to see the ports in use, and to see their status.

QUESTION 1 Based on the outputs, which switch is the current root bridge?

QUESTION 2 What is the current bridge priority of the root bridge?

Before proceeding, go to the "Lab 5 Part 2 - Mitigating Against STP Manipulation & CAM Table Attacks - QUESTIONS" quiz on the Brightspace page and enter your answers for questions 1 & 2. Leave the quiz open while you complete the rest of the lab sheet.

Step 2: Assign Core as the primary root bridge.

Using the **spanning-tree vlan 1 root primary** command, and assign **Core** as the root bridge.

Step 3: Assign SW-2 as a secondary root bridge.

Assign **SW-2** as the secondary root bridge using the **spanning-tree vlan 1 root secondary** command.

Step 4: Verify the spanning-tree configuration.

Issue the **show spanning-tree** command to verify that **Core** is the root bridge.

```
Core# show spanning-tree
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 24577
Address 00D0.D31C.634C
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

QUESTION 3 What is the bridge priority of the new root bridge?

QUESTION 4 What is the current bridge priority of the secondary root bridge?

Before proceeding, return to the quiz and enter your answers for questions 3 & 4. Leave the quiz open while you complete the rest of the lab sheet.

Part 2: Protect Against STP Attacks

Secure the STP parameters to prevent STP manipulation attacks.

Step 1: Enable PortFast on all access ports.

PortFast is configured on access ports that connect to a single workstation or server to enable them to become active more quickly. On the connected access ports of the **SW-A** and **SW-B**, use the **spanning-tree portfast** command.

Step 2: Enable BPDU guard on all access ports.

BPDU guard is a feature that can help prevent rogue switches and spoofing on access ports. Enable BPDU guard on **SW-A** and **SW-B** access ports.

Note: For marking purposes in this lab, please enable it on each individual port (rather than enabling it globally on all ports).

Part 3: Configure Port Security and Disable Unused Ports

Step 1: Configure basic port security on all ports connected to host devices.

This procedure should be performed on all access ports on **SW-A** and **SW-B**.

a. Enable port security on all access ports on SW-A and SW-B.

   Note: A switch port must be configured as an access port to enable port security.

b. Set the maximum number of learned MAC addresses to **4**.

c. Allow the dynamically learned MAC addresses to be automatically saved (i.e. to "*stick*") in the running configuration

d. Set the violation mode to **restrict**..

Step 2: Verify port security.

a. On **SW-A**, issue the command **show port-security interface f0/1** to verify that port security has been configured.

```
SW-A# show port-security interface f0/1
Port Security : Enabled
Port Status : Secure-up
Violation Mode : Restrict
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 4
Total MAC Addresses : 0
Configured MAC Addresses : 0
Sticky MAC Addresses : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```

b. Ping from **C1** to **C2** and issue the command **show port-security interface f0/1** again to verify that the switch has learned the MAC address for **C1**.

QUESTION 5 What is the "Total MAC Addresses" value in the output after doing the ping?

Before proceeding, return to the quiz and enter your answer for question 5. Leave the quiz open while you complete the rest of the lab sheet.

Step 3: Disable unused ports.

Disable all ports that are currently unused.

If you have correctly configured all parts of the lab your activity score should now be showing as 100%. If so, click on "check results" in the activity window. Return to the Brightspace quiz one last time and enter the code into the appropriate question box (Q6) of the quiz.

You have completed the lab – please submit the Brightspace quiz.